Check for updates

The Meteoritical Society

Meteoritics & Planetary Science 55, Nr 2, 274–293 (2020)
doi: 10.1111/maps.13431

Determining the age and possibility for an extraterrestrial impact formation mechanism of the Ilumetsa structures (Estonia)

A. LOSIAK[1,2]*, A. JÕELEHT[3], J. PLADO[3], M. SZYSZKA[4], K. KIRSIMÄE[3], E. M. WILD[5], P. STEIER[5], C. M. BELCHER[2], A. M. JAZWA[1], and R. HELDE[3]

[1]Planetary Geology Lab, Institute of Geological Sciences, Polish Academy of Sciences, Warsaw, Poland
[2]wildFIRE Lab, Hatherly Laboratories, University of Exeter, Exeter, UK
[3]Department of Geology, University of Tartu, Tartu, Estonia
[4]Institute of Geology, Adam Mickiewicz University in Poznań, Poznań, Poland
[5]VERA Laboratory, Faculty of Physics—Isotope Physics, University of Vienna, Vienna, Austria
*Corresponding author. E-mail: anna.losiak@twarda.pan.pl

(Received 13 October 2018; revision accepted 22 October 2019)

Abstract–The Ilumetsa site, in Estonia, consists of two round, rimmed structures that are 725 m apart. The structures are listed as proven impact craters in the Impact Earth database, despite lack of commonly accepted, unequivocal proof of extraterrestrial collision identified at this location. We excavated trenches though the Ilumetsa Large and Ilumetsa Small structures and found small pieces of charcoal within the putative proximal ejecta in both structures, in a similar geological setting as previously identified charcoal in Kaali (Losiak et al. 2016) and Morasko craters (see Szokaluk et al. 2019). Our ^{14}C dating of charcoal allowed us to conclude that these crater-like features formed simultaneously between 7170 and 7000 cal. years BP, about 7 ka after deglaciation of this area. A ground penetrating radar survey of the nearby bog shows that no additional Ilumetsa structures bigger than 40 m exist. Geochemical studies of the ejecta and a search using a metal detector did not reveal any clear indication of extraterrestrial material. This suggests Ilumetsa may have been formed by an impact of stony-iron or stony body, which got significantly weathered in a wet-temperate climate. The mystery of the formation of the structures at Ilumetsa remains; however, due to significant circumstantial evidence discussed herein, we are confident to call it a "probable" impact site.

INTRODUCTION

The Ilumetsa site in SE Estonia consists of two structures, located 725 m from each other. Their diameters are 75–80 m (Põrguhaud/Hell's Grave/Ilumetsa Large—IL) and ~50 m (Sügavhaud/The Deep Grave/Ilumetsa Small—IS) with true depths of about 8 and 3.5 m, respectively (Plado 2012; Fig. 1). Both structures are surrounded by a rim up to a few meters high. The rims are highest in their eastern parts with maximum rim heights of 4.5 (IL) and 1.5 m (IS). The Ilumetsa site and associated craters are listed as a proven meteorite impact in the Impact Earth database (Osinski and Grieve 2019), but neither remnants of the projectile nor other identification criteria, such as planar deformation features (PDFs), have been found up to this point (Plado 2012). The presence of deformed sedimentary beds (including the rim consisting of sands mixed with numerous clayey till lenses; Aaloe 1963) and only a small thickness of glacial sediments favor an impact rather than a glacial (e.g., kettle- or sinkhole) origin for the craters (Plado 2012).

The Large Ilumetsa structure (IL) has previously been dated by undertaking radiocarbon analyses of gyttja and peat from the very bottom of the crater-shaped depression (Liiva et al. 1979). The lowermost organic beds were dated to 6030 ± 100 ^{14}C years (7170–6660 cal. years BP; recalibrated with IntCal13). Raukas et al. (2001) proposed that the structures were approximately 600 years older in age (6542 ± 50 ^{14}C years or 7570–7320 cal. years BP; IntCal13) by analyzing peat that contained glassy spherules found at a depth of

© 2020 The Authors. Meteoritics & Planetary Science published by Wiley Periodicals, Inc. on behalf of The Meteoritical Society (MET)
This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

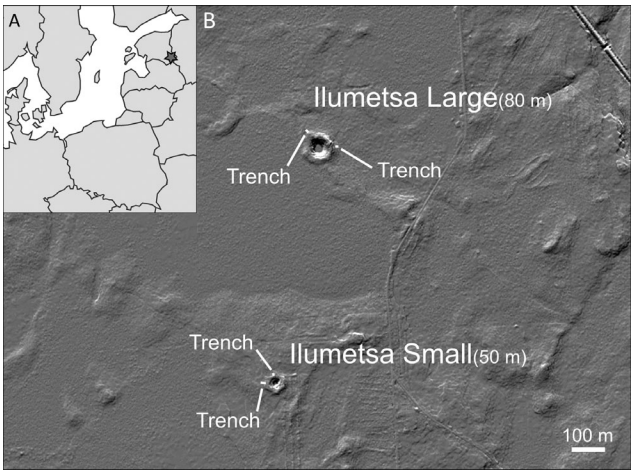


Fig. 1. A) Location of the Ilumetsa structures on a regional map. B) Shaded relief map showing areas where two Ilumetsa structures are located, based on LiDAR digital elevation model data and location of the trenches (lengths of trenches are up to scale). The 4.0 m long trench at the Ilumetsa starts at the 57°57′37.28″N and 27°24′12.59″E, the 8.0 m long trench at the Ilumetsa Small starts at the 57°57′12.60″N and 27°24′00.99″E, and the shorter, 4.2 m long trench started at the 57°57′12.99″N and 27°23′59.29″E. A large flat area south from Ilumetsa Large is a Põrguhavva bog. Data from Estonian Land Board.

5.7 m from the Meenikunno bog that is ~6 km SWW from the Ilumetsa structures. These spherules are reported as being up to a few millimeters in diameter and have been interpreted as dissipated melt or condensed vapor. However, their chemical composition (and thus, the presence of extraterrestrial material as relating them to the Ilumetsa structures) was not reported. The smaller structure (IS) has not been previously dated itself owing to the absence of organic-rich sediments suitable for dating. Consequently, the temporal relationship between IL and IS has not yet been established.

The aim of this study is to: First, assess the ages of both Ilumetsa structures by ^{14}C dating of organic material covered by what we believe to be a layer of proximal ejecta. In order to achieve this, we have searched for possible impact-related charcoals because this approach has recently been successfully applied to the dating of three other craters found in Estonia; the Kaali Main crater (Losiak et al. 2016), the Kaali 2/8 crater (Losiak et al. 2018), and the Morasko ejecta blanket (Szokaluk et al. 2019). Second, we attempt to determine if the Ilumetsa structures were indeed created by a collision of an extraterrestrial body with the Earth surface by means of looking for meteorite fragments in the field and for geochemical traces of extraterrestrial material in the proposed proximal ejecta, and by searching for additional structures that may be related to this potential strewn field. As such we aim to combine a range of approaches to assess whether or not these structures can be related to a small impact event.

MATERIALS AND METHODS

Geological Setting

Geologically, the Estonian bedrock is composed of early Paleozoic sediments overlying Precambrian crystalline basement. Paleozoic sediments have a shallow (0.1–0.3°) dip to the south and are exposed below Quaternary sediments in approximately latitudinal zones becoming successively younger moving southward from Ediacaran–Cambrian strata in the north to Devonian strata in the south. The Ilumetsa "target rocks" are of Middle Devonian age and are of very weakly cemented light-yellow to reddish sandstones (containing some silt interlayers). These are overlain by up to ~2.5 m thick layer of brown Quaternary basal till and/or yellowish fluvioglacial sand locally covered by Holocene peat. Near the Ilumetsa structures, the Quaternary cover consists of brownish sandy till that is <1 m in thickness. The region was deglaciated ~14 ka ago (Stroeven et al. 2016).

In the area of Ilumetsa, the rocks directly beneath the Quaternary sedimentary cover are sandstones of the uppermost Middle Devonian Burtnieki Regional Stage (Fig. 2; Raukas and Teedumäe 1997). The sandstones were deposited in shallow-marine deltaic environment and are comprised of fine-grained sand (50–70% of the rock) and a smaller percentage of the medium-grained sand (10–30% of the rock), while the percentage of the very fine-grained component ranges 6–20%. The share of other fractions rarely exceeds 5%. The typical thickness of these south-dipping cross-bedded sandstone sets is 20–30 cm. The sandstones include ~10–15 cm thick lenses of reddish-brown siltstones and clays (see red silty-clay layer on Fig. 3b). Mineralogically, rocks of the Burtnieki Formation (Raukas and Teedumäe 1997) are predominantly quartzose and feldspathic arenites with quartz content typically of 70–90% of the total rock. Micaceous arenites (content of micas up to 50%) occur as rare, thin interbeds. The heavy fraction is dominated by ilmenite (45–65%). The share of allothigenic transparent minerals in the heavy fraction is ~15–30%. It is made of zircon (40–70%), tourmaline (7–20%), and staurolite (3–15%) with some kyanite.

Sampling Protocol and Samples Taken

Trenches were excavated into the proposed proximal ejecta in a radial pattern around the Ilumetsa structures (see Fig. 1 for locations) in order to search

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Fig. 2. An outcrop showing the typical, undisturbed stratigraphy in the region of Ilumetsa. The image was taken in Taevaskoja, 30 km to the north–west from the Ilumetsa site. The ~20 m high wall of pinkish and whitish, cross-bedded sandstone of Middle Devonian aged Burtnieki sandstone is overlain by glacial and fluvio-glacial deposits overgrown by forest. Locally, including at the Ilumetsa site, the Burtnieki sandstone is significantly less well lithified than shown here. Image: Tõnu Pani, October 2008. (Color figure can be viewed at wileyonlinelibrary.com.)

for impact-related charcoals. At IL structure one trench, which was 4 m long, located at about 16–20 m from the rim at the NE side of the structure (57°57′37.28″N, 27°24′12.59″E) was studied. The ejecta blanket visible in the trench consisted mostly of massive, fine- to medium-grained sand with rare clasts of rounded crystalline rocks intermixed with randomly distributed lenses of red, medium-grained sand (Fig. 3). At a depth of 130 cm (at the rim end of the trench), there was a nearly horizontal, about 10 cm thick, occasionally 20 cm thick, layer of reddish silty-clay characteristic of the undisturbed Burtnieki Formation sandstones. At the surface, a podzol soil with a well-developed bleached eluvial horizon at the depth of 10–30 cm was present.

A 10–20 cm thick lens-shaped zone was identified at around 55–65 cm depth within the massive sand. All charcoal samples (except ones from the organic matter horizon of the soil taken for reference) came from this zone (Fig. 3). This charcoal-rich zone has the same sediment structure and texture and seems to differ only by the addition of charcoal pieces. The sampled lens occurs at the distance between 1.7 and 2.8 m from the edge of the crater-shaped features rim (here and below the distances are given from the trench edge, furthest from the crater-shaped feature). The longer axis of the lens was slightly inclined in the outward direction. Most pieces of charcoal are 1–2 mm in length, but a few larger pieces (up to 1 cm) were also identified. Despite a thorough search, no pieces of charcoal were found in the first two meters of the trench (farthest away from the rim). The charcoals do not appear to have been moved from the soil profile down by biological action and it would be impossible for charcoals that are up to 1 cm in size to be transported down the profile by leaching. Similarly, no charcoal below the O horizon (at depths of >40 cm) was found in any of the small, shallow outcrops that we exposed in a >100 m distance from the rim of IL and IS in order to observe the undisturbed target rocks. As such the charcoal seems to be contained only with these sandy lenses within the proposed ejecta horizons.

Two further shallow trenches were excavated around 70 cm in depth (this depth was limited by the ground water level) in order to investigate the nature of the undisturbed sediments. They were located 10 and 15 m to the east from the outermost site of the trench. Those locations were selected based on the angle of the slope at the structure's rim. These trenches revealed the presence of a bleached podzol horizon developed on an undisturbed yellowish-beige sand containing dark, soft spots ~1 × 1 cm in size that are suspected to be Mn-oxyhydroxide concretions.

Two trenches were excavated at the IS structure; located at a ~90° radial angle from each other (Fig. 1). The first was a longer (8.0 m) trench up to ~150 cm deep, located 57°57′12.60″N 27°24′00.99″E on the SSE side of the crater-shaped feature, within a section characterized by the steepest outer slope (Fig. 4). A second shorter (4.2 m) trench up to ~120 cm in depth, located 57°57′12.99″N 27°23′59.29″E was excavated at the SW side, which had a shallower outer slope (Fig. 4). Both trenches ran through massive, medium-grained sand that contained lenses/undulating layers of gravel and coarse sand. In the deepest part of the longer trench (at depths >140 cm), Devonian-aged reddish fine-grained, undisturbed sandstone-siltstone was exposed. This same layer was also visible beneath the soil level in the section of the trench farthest from the rim. This section of the trench was located outside the proposed ejecta sequence (judging from the slope angle). In the middle part of the section at a depth 60–80 cm, irregular clayey lenses of till-like material occurred. Occasionally, horizontal red staining was present, likely caused by oxidation and related to water table fluctuations.

The shorter trench appeared to contain fewer charcoals, there was no obvious charcoal-rich lens like that found in IL. A total of six charcoal samples were collected and were at least 5 mm in length (see Fig. 5, Ilume_S_2_3). The charcoals were found to occur at approximately the same depth to each other, but did not appear to form a continuous layer.

In the longer trench, there was significantly more charcoal present compared to the shorter trench and the

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

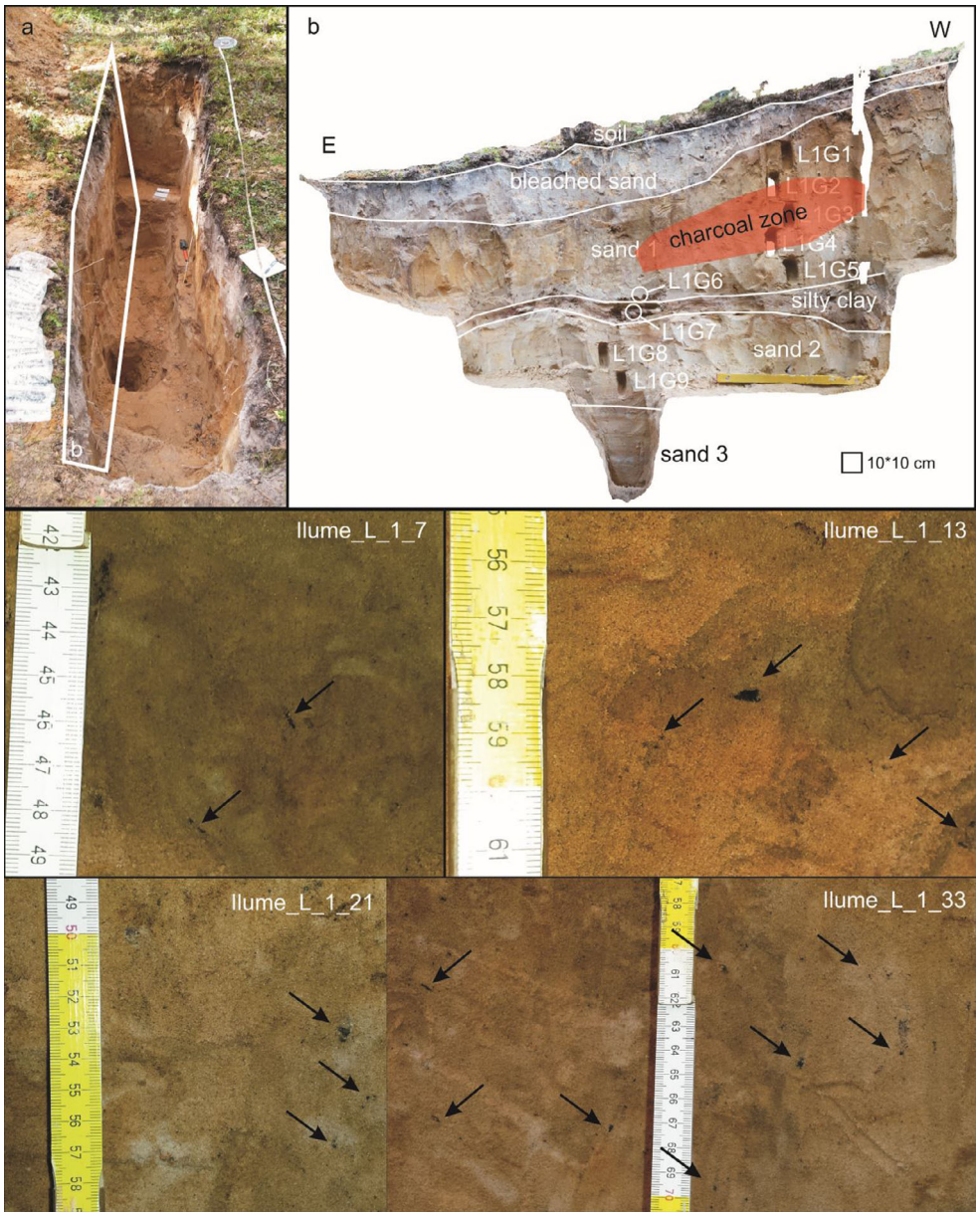


Fig. 3. A trench dug through the proximal ejecta blanket of Ilumetsa Large (4 m long and up to 180 cm deep). Coordinates of the eastern end are 57°57′ 37.28″N and 27°24′12.59″E. a) A view of the full trench in the direction toward the feature from east to west. Distance from the far end of the trench to the rim is ~16 m. b) A side view of the trench. A bleached eluvial zone of podzol is located just below the O level. Below there is a wedge of well-sorted, fine- to mid-grained sand (sand 1; the same grain size and sorting as in the bleached). A semi-horizontal layer of red, plastic silty clay at 100–140 cm depth is ~10 cm, locally >20 cm, thick. Below the clay level, there is beige to orange fine- to mid-grained sand with undulating dark red silty clay layers (sand 2), and in the lowest part of the section opened in trench (sand 3) sand is darker beige, medium-grained, and without red lenses, there are also some signs of sedimentary bedding. L1G1–L1G9 denote samples for chemical analysis. Photographs below show charcoal samples selected for ^{14}C dating (adjacent soil samples for chemical analysis). Charcoal was found within the ~20 cm thick and 1.1 m long zone (marked in red on the section [b] of this figure). This lens continued in the opposite wall of the trench. (Color figure can be viewed at wileyonlinelibrary.com.)

distribution of the charcoals within the strata was more complex. Charcoal pieces were not distributed along the entire trench, but between the distance of 4.9–7.0 m (with a couple of pieces at 8.0 m distance; distances are counted from the beginning of the trench located the farthest away from the rim). Most of the charcoal pieces, which were typically <5 mm diameter, were found within a 20 cm diameter lens-like gray zone of massive silty sand

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

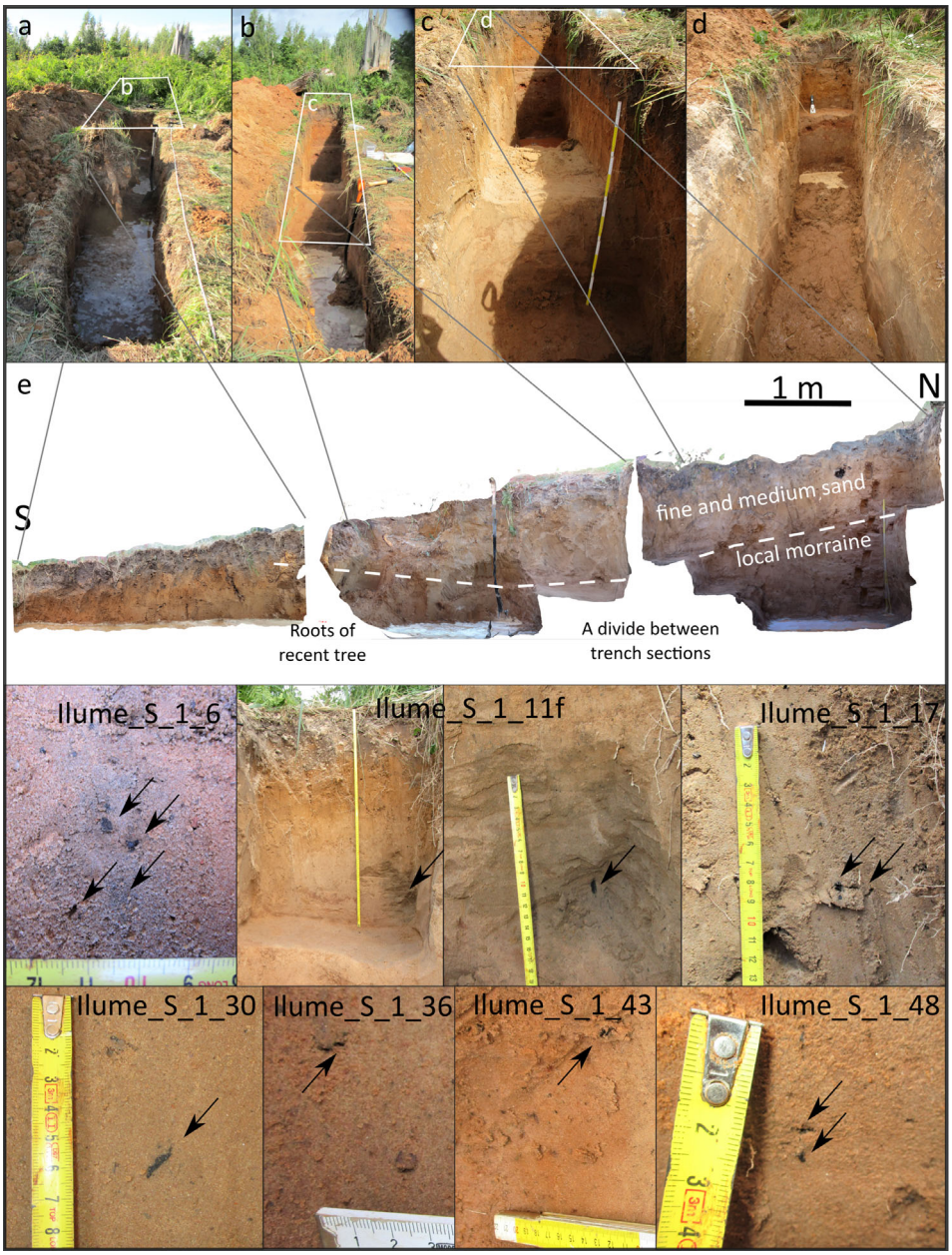


Fig. 4. A trench dug through the proximal ejecta blanket of Ilumetsa Small (8 m long, up to ~150 cm in depth where water level was present; coordinates of the southern end are 57°57′12.60″N and 27°24′00.99″E). a–d) Photo of sections of the trench looking in the direction of the structure (tree tops visible in the background are growing inside the crater-shaped feature). Please note groundwater level made it impossible to dig deeper. The "step" between second and third section of the outcrop and visible on image (b) and (c) was subsequently removed. The section visible on photo (d) was mated later depending on the groundwater level. e) A side view of the trench where two layers with different properties are visible (the approximate boundary between them is marked with a thin white line). Similar to the shorter outcrop made in the same structure, the soil is developed on the orange fine- to medium-grained sand containing lenses of material with gravel-sized, well-rounded pebbles made of crystalline rocks. This layer includes numerous, semi-horizontal dark-red oxidation patches (e.g., as one visible on the image Ilume_S_1_43 to the left from the charcoal). Deeper than ~80 to 100 cm, there is beige sandy local till material with patches of clay and cm-sized crystalline pebbles. Charcoal was found in five different locations within this trench, Ilume_S_2_3 is shown here. Charcoal was not forming a clear but discontinuous layer, like it was visible in other locations (Ilumetsa Large—this study, Kaali Main crater [Losiak et al. 2016], Kaali 2/8 crater [unpublished field study by the same authors]). (Color figure can be viewed at wileyonlinelibrary.com.)

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

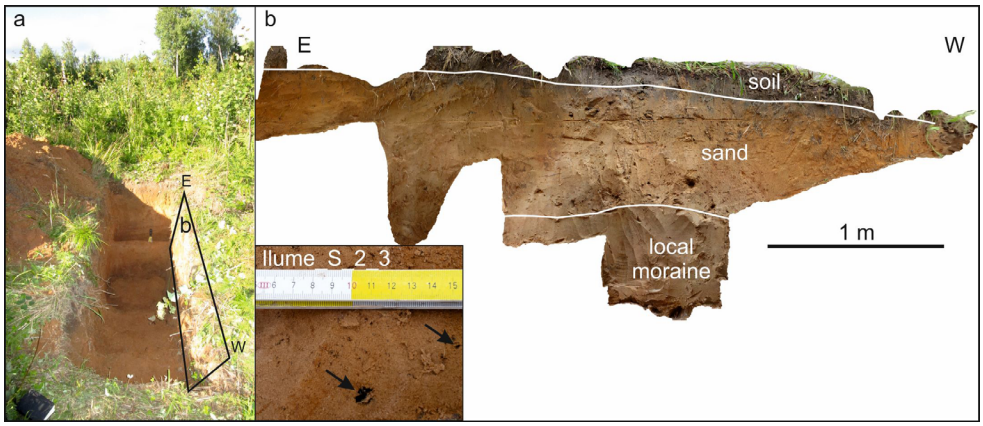


Fig. 5. A trench dug through the proximal ejecta blanket of Ilumetsa Small (4.2 m long, up to ~120 cm in depth). Coordinates of the western end are 57°57′12.99″N and 27°23′59.29″E. The trench was located at the shallow gradient side of the structures rim. a) A view of the full trench. The distance from the eastern end of the trench to the rim is ~9 m. b) A side view of the trench. Soil is developed on the layer of orange fine- to medium-grained sand containing ~5% of gravel sized, well-rounded pebbles made of crystalline rocks. This layer includes numerous, semi-horizontal dark-red oxidation patches. All charcoal pieces were located in this layer (including Ilume_S_2_3 that was ^{14}C dated). Starting from 90 to 100 cm below surface, there is a beige sandy local till material with patches of clay and cm-sized crystalline pebbles. (Color figure can be viewed at wileyonlinelibrary.com.)

located at the depth of 65–85 cm. A large piece of charcoal 2.5 cm in length (Ilume_S_1_11g) was found in the middle of this zone. A second similar, but smaller in size, lens-like charcoal zone was identified around 50 cm away from the first one. In addition, a couple of pieces of charcoal were found just above the level of ground water (at the depth of 90–110 cm) and there were other charcoal particles (Ilume_S_1_48) at shallower depths of ~50 cm below the soil horizon.

Assessment of Subsurface Features

A ground penetrating radar (GPR) was used to survey the Süvahavva bog, located between the IL and IS structures, in order to determine the possible existence of any additional smaller crater-like features that might have been buried under peat after the impact. A radar device Zond 12e by Radar Systems Inc. was used to map peat thickness and the sub-peat topography. GPR profiling was performed using frequency 300 MHz. Measurements were triggered by an odometer wheel at 0.1 m spacing and location of profiles was recorded using instrument attached to a GPS device (estimated accuracy in forested area better than 10 m). Recording time was 500 ns. Data were processed and interpreted with Prism2 software. The sub-peat reflection two-way time was converted to depth using a relative permittivity value 70 that commonly holds for boreal bogs (Plado et al. 2011; Hiiemaa et al. 2014; Habicht et al. 2017; Sibul et al. 2017). The thickness of peat and peat/sub-peat sediment types determined by the GPR was checked at five locations using a hand drill.

Searching for Meteoritic Material

In order to find evidence of an impactor an extensive search for any pieces of meteorite was performed using a Lorenz Deepmax z1 metal detector. A Pulse ground balancing system in this device sends out a short and intensive (primary) electromagnetic impulse into the ground, the magnetic components of these pulses produce eddy currents in conducting materials such as metal objects. The eddy currents produce a secondary electromagnetic field that gradually dies away after the electromagnetic impulse has been turned off. If conductive material is present, this increases the time taken for the field to return to zero what can be detected. used in this study can also indicate the type of metal (or rather its conductivity). This allows discrimination between a ferrous versus non-ferrous metal object without the need for digging. In some cases (e.g., in the Kaali craters in Estonia), after testing the response of the device in the particular area for different types of objects, it was also possible to distinguish iron man-made objects (e.g., pieces of farming equipment) from meteorites. This is because the presence of Ni influences electrical conductivity. The approximate detection abilities of the device, for example, would allow the detection of a 20 L metal fuel tank at depth of ~3 m (larger objects can be located deeper, smaller objects have to be located shallower).

In order to find magnetic particles, a dry sample from the charcoal-containing zone within the 4 m trench excavated in the IL structure was screened with a

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

magnet. In addition, the surroundings of the structures were surveyed with a metal detector and a few highly oxidized magnetic particles were found. These particles and the magnetic fraction from the screening of the sample were embedded in resin and polished stepwise and analyzed under a scanning electron microscope with energy dispersive X-ray spectroscopy (SEM EDS) at every polishing step using a ZEISS EVO MA15 scanning electron microscope with an Oxford AZTEC-MAX energy-dispersive detector at the University of Tartu.

The composition of selected major and trace elements in samples taken from sections of the trenches was determined using inductively coupled plasma mass spectrometry and optical emission spectrometry (ICP-MS/ICP-OES) in multi-acid digested (HNO_3, $HClO_4$, HF, HCl) at the Bureau Veritas Commodities Canada Ltd, Vancouver, Canada. Platinum group elements (PGE) in selected samples were analyzed by instrumental neutron activation analysis (INAA) following the nickel sulfide fire assay at the Activation Laboratories Ltd., Canada.

Age Estimates—Radiocarbon Dating

Charcoal samples were sent for radiocarbon dating to the Vienna Environmental Research Accelerator (VERA) laboratory at the University of Vienna (Austria). The samples were chemically pretreated with the standard ABA (acid–base–acid) procedure used at VERA (Wild et al. 2013). The ^{14}C measurements were performed as described in Steier et al. (2004). The uncalibrated/conventional ^{14}C ages, given in ^{14}C years BP (before present = before 1950 AD), are calculated following conventions summarized, in Stuiver and Polach (1977). The uncalibrated ^{14}C ages are transformed into calendar time ranges (given in cal. BP) to account for variations of the past atmospheric $^{14}C/^{12}C$ value (and the Libby half-life used for the calculation of the uncalibrated ^{14}C ages) using the IntCal13 atmospheric curve (Reimer et al. 2013) and the calibration program OxCal v4.3.2. (Ramsey and Lee 2013). All calibrated ages were rounded to 10 yr in the OxCal program.

Charcoal samples for ^{14}C dating were selected in order to assess the potential linkage to formation by a small extraterrestrial collision according to approaches previously used by Losiak et al. (2016, 2018). Only samples that were associated with good quality field photos and were present in large enough quantity for ^{14}C, reflectance measurements, and SEM analysis were taken into consideration. Charcoals from the IL trench were, based on field analysis, estimated to be similar in age because all of the particles were found close to each other (at least one particle every 5 cm) within a consistent 1.5 m long and 15–20 cm thick layer. In order to check for any variability in age within this apparently uniform layer, four samples were selected from different ends of the charcoal-bearing zone.

From the shorter Ilumetsa Small trench, we selected one sample, because only a couple of particles were found directly close to one another. However, the charcoal distribution was more complex in the longer trench excavated in the Ilumetsa Small structure. Even in the field, we expected that these probably were formed by different events at different times. In order to test hypotheses on their genesis, we determined ^{14}C age of eight samples based on the following sampling strategy. The "Ilume_S_1_6" charcoal sample was selected because it was found far from the other particles; this piece was also farther away from the rim than expected based on at the IL structure. This charcoal sample was located within a grayish zone that appeared to be a closed burrow tunnel (Zazula et al. 2007). Therefore, we suspected that it may be an example of charcoal formed some time after the formation of the crater structure that was dragged underground by biological action. The "Ilume_S_1_48" sample was also located away from other charcoals, at a shallow depth (50 cm depth while most of other charcoals were found at the depth of ~80 to 90 cm even farther away from the rim). We hypothesized that this charcoal was part of the deluvial soil sequence that had initially been located on the surface, and was later covered by 50 cm of soil that came from the erosion of the rim such that it was covered relatively quickly after the formation of the structure. The "Ilume_S_1_43" sample was selected because it was located much deeper than other samples (115 cm). It was found close to the ground water level. Samples "Ilume_S_1_11g," "Ilume_S_1_17," and "Ilume_S_1_30" came from different areas within the grayish lens that were observed to be rich in organic material and charcoal (Fig. 4, Ilume_S_1_11g). We hypothesized that these charcoal samples may represent reworked charcoals formed during a wildfire(s) that occurred prior to formation of the structure. We assumed that samples Ilume_S_1_32 and Ilume_S_1_36 were the most likely to be derived themselves from an extraterrestrial collision during the formation of the structures.

RESULTS

Ground penetrating radar profiling revealed that the thickness of peat in Süvahavva bog is mostly in the range of 1.0–2.5 m, although slightly thicker toward the center of the bog (Fig. 6). The lower part of the organic sequence is formed by fen peat whereas the uppermost

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

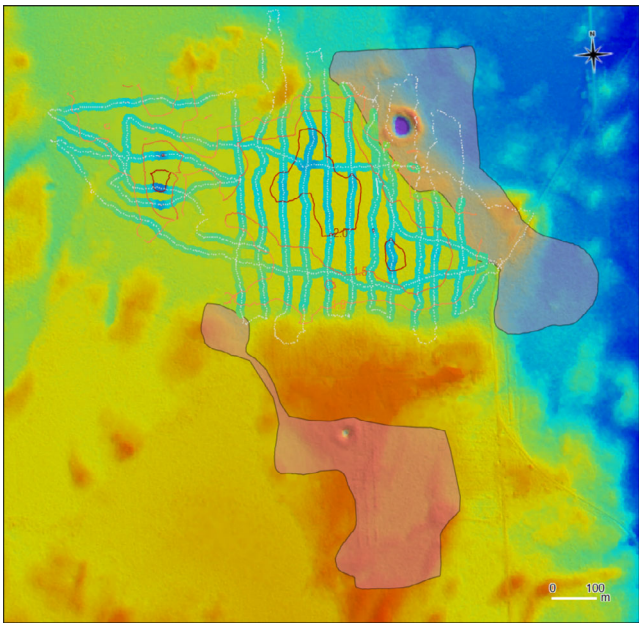


Fig. 6. The topographic map of the Ilumetsa area showing GPR tracks (white dots) and area covered by the metal detector screening (pink zone with black outline). Sub-peat topography is shown in color around GPR profiles (white dots). Both sub-peat topography and surface topography have the same color scale (blue—50 m a.s.l. to orange—56 m a.s.l.). Peat thickness isolines of 1.0, 1.5, and 2.0 m are given with orange lines. The areas checked with metal detector for the existence of iron-bearing meteorites are marked with pink. Most of the pink areas were fully covered by metal detector tracks, except locally, when small ponds or fallen trees were encountered. (Color figure can be viewed at wileyonlinelibrary.com.)

part (<0.8 m thick) of the profile is sphagnum dominated. The sub-peat topography on the GPR profiles is smooth and no structures that could be interpreted as part of rim or crater cavity were detected. An N–S elongated topographic feature is present in the western part of the studied bog beneath the peat (red isolines on Fig. 6). The complex shape of this feature in plan view and its shallow depth excludes it as having an impact origin. It appears to resemble similar topographic features in the region that are glacial in origin.

Despite the 1.5 day-long search with a metal detector over ~270,000 m^2 of the area, around both structures, no clear signs of meteoritic material were found (Fig. 6). Numerous pieces of farming equipment (especially horseshoes) and parts of military equipment were found. Additionally, areas were identified that contained high amounts of bog iron that includes magnetite. In only one location, on the NW rim of the Ilumetsa Small, we identified a location that was responding to the metal detector at frequency characteristic to iron-nickel alloy, similar to that found around the Kaali craters (M. Szyszko during 2017 field campaign). Therefore, despite no large meteoritic body being found, a number of small weathered pieces that adhered to the small hand magnet were collected.

SEM-EDS analysis revealed that most of the Fe-rich fragments collected in the field and during additional screening in the lab are aggregates of Fe-oxyhydroxide precipitates. These appeared to grow concentrically in distinct layers with the composition of the layers varying between goethite and hematite (Fig. 7). Most of the analyzed material did not include any Ni (at detection level <0.1 wt%), but we encountered four grains with high Ni content. These were three small 10–20 μm roundish grains and a single 50 μm long piece. The Ni content in the four analyzed grains is consistently at (based on EDS spectra) 86–89 wt%.

Bulk sediment geochemical analyses (Table 1) of samples taken from the trench at IL show apparent Ni and Cr enrichment (15 and 44 ppm, respectively) in sample L1G7 from the 0.95 to 1.05 m depth interval. There is no enrichment evident in Ni/Ti ratios suggesting that the elevated Ni, Cr, and other metal concentrations are most likely due to sedimentological variation as they appear to correspond with a silty clay layer in this interval (Fig. 8a). No Ni or Cr enrichment was observed in the depth interval where the charcoal fragments were found.

Similarly, in the profile at Ilumetsa Small, there are two spikes in both Ni and Cr content at depths 0.85–1.15 m and 1.40–1.50 m, but the enrichment is diminished in Ni/Ti ratio (Fig. 8b). Similar to the IL section, the changes in trace element concentrations seem to be related to the appearance of more clay-rich sediments at these depth intervals whereas the deepest sample is from the undisturbed Devonian basement. There is no trace element enrichment in the interval with the gray silty clay lens within which the charcoal particles were found.

Instrumental neutron activation analysis, undertaken for identifying PGE in samples from the IL profile (samples L1G1–L1G9) and the Devonian siltstone-sandstone from Ilumetsa Small, did not show any PGE enrichment, and Ir content remained below the detection limit (0.1 ppb).

We dated 13 charcoal samples from both Ilumetsa structures: four samples from IL, one sample from the short trench, and eight samples from the long trench at IS (Table 2, Figs. 3–5). The seven samples from both the crater-like features form a consistent age group and span the time interval from 7670 up to 6950 cal. BP, which is slightly older, but consistent with the previous Liiva et al. (1979) dates of gyttja and peat from the IL structure.

Six dated samples do not fit within the age group above. Three samples (Ilume_S_1_6, Ilume_S_1_43, and Ilume_S_1_48) from the longer trench of IS are younger

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

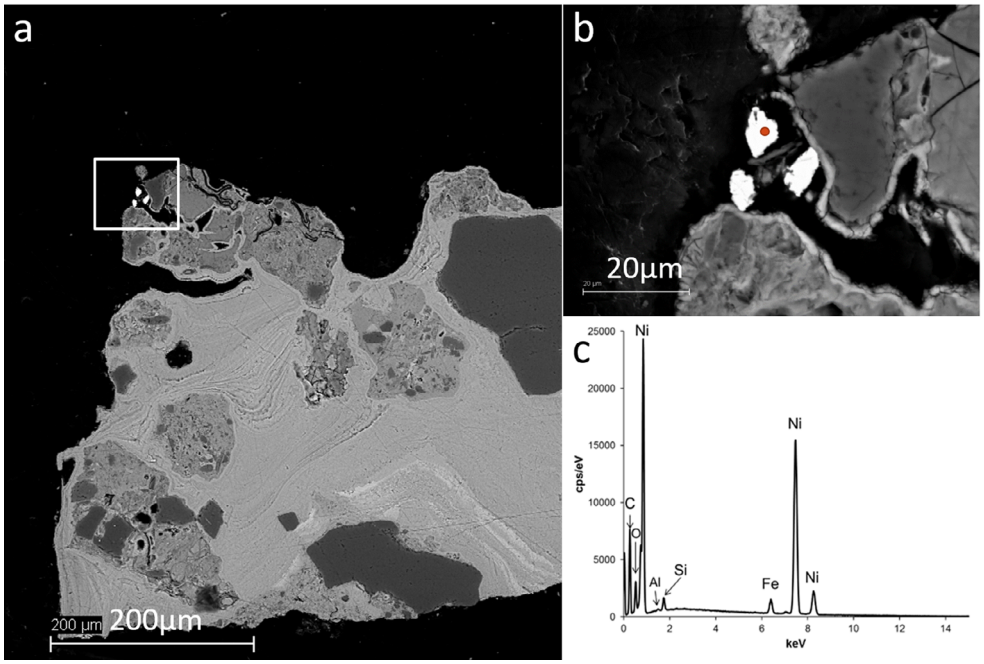


Fig. 7. Ni-rich particles within Fe-oxides found on the NW rim of the Ilumetsa Small. a) Fe-oxide grain. White rectangle shows location of section (b) of this figure. b) A close-up of the area where Ni-rich particles are present. The dot on a bright grain shows location of the EDS analysis point. c) EDS spectra of the Ni-rich particle. (Color figure can be viewed at wileyonlinelibrary.com.)

than the Liiva et al. (1979) age. They do not form any consistent age group and are younger than the main age group by 1000 years to over 5000 years. Three samples (Ilume_S_1_11g, Ilume_S_1_17, Ilume_S_30) are older than the samples from the main age group by over 1000 years.

DISCUSSION

Search for Other Crater-Like Structures in the Area of Ilumetsa

Ground-penetrating radar profiling did not reveal any topographic features in the Süvahavva bog that could be interpreted as an additional Ilumetsa crater feature. However, we cannot exclude that small structures may not have been detected in our survey that had a 50 m average spacing between the profiles. It is possible that a structure could have remained unrecognized between the 50 m profiles such if the outer flank of the features rim was low enough. The diameter of the IL and IS structures at recognizable rim foothill level is, respectively, about 100 m and 60–65 m, that is, about 20% larger than the rim-to-rim diameter. Therefore, if there are additional structures, the rim-to-rim diameter of these within Süvahavva bog must be smaller than 40 m in diameter.

Additionally, the hypothetical crater must be completely buried under the peat, that is, the thickness of peat must be greater than the rim height. Pike (1977) empirically related lunar crater rim diameter D_r (km) to rim height R_e (km) using equation $R_e = 0.036\ D_r^{1.014}$. Melosh (1989) confirmed that this relationship holds also for both impact and explosion craters on Earth. It also holds for IL and IS if local topographic variations and some erosion of the rim are taken into account. This equation suggests that a crater with diameter of 40 m should have a rim 1.4 m high, whereas the peat thickness in Süvahavva bog varies between 1.25 and 2.5 m.

Based on our GPR analysis, we can reject a hypothesis that there is an additional Ilumetsa structure >40 m within the Süvahavva bog. However, despite our dense profile spacing, we cannot exclude existence of a peat-buried crater of <40 m in diameter in the Süvahavva bog.

Lack of Clear Signs of Extraterrestrial Material

During our 1.5 day field campaign devoted to looking for meteorites in 2017, we found no iron meteorites in the area surrounding the Ilumetsa structures (Fig. 5). Previous search campaigns (using magnetometers) were also unsuccessful (Aaloe 1963;

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Geochemical composition of samples from profiles in trenches at Ilumetsa Large and Ilumetsa Small.

Sample		Depth (m)	Al	Fe	Ti	Ca	Mg	Na	K	P	Cr	Ni	V	Co	Mn	Zr	Rb	Sr	Ba	U	Th	Mo	Cu	Pb	Zn	As
			wt%									ppm														
L1G1	Ilumetsa	0.20–0.40	0.86	1.4	0.09	0.05	0.03	0.099	0.65	0.012	9	2	6	1.4	266	39.7	17.9	22	129	0.4	1.8	0.4	17.1	5.3	15	1
L1G2	Large	0.40–0.55	0.83	0.97	0.085	0.04	0.02	0.067	0.65	0.014	7	2.1	6	1.1	260	33.8	19.6	19	131	0.4	1.5	0.3	14.5	4.4	19	1
L1G3[a]		0.55–0.70	0.76	0.87	0.081	0.03	0.01	0.073	0.6	0.012	7	1.7	5	1.1	149	30.8	16.6	19	122	0.4	2.1	0.2	9.8	4.4	20	<1
L1G4		0.70–0.85	0.70	0.68	0.076	0.03	0.02	0.06	0.61	0.006	7	1.2	5	0.7	100	27.2	16.3	17	114	0.4	1.6	0.2	4.0	3.4	11	<1
L1G5		0.85–1.00	1.09	0.87	0.103	0.05	0.05	0.102	0.74	0.014	8	3.4	8	1.8	135	38.6	22.7	25	154	0.5	3.0	0.2	6.7	5.0	18	<1
L1G6		0.90–0.95	1.65	1.22	0.099	0.10	0.07	0.197	1.34	0.009	9	3.0	8	2.1	218	54.0	38.5	43	263	0.7	3.2	0.3	6.3	8.2	8	2
L1G7		0.95–1.05	4.83	5.12	0.29	0.26	0.42	0.504	2.28	0.026	44	15.4	54	8.8	551	149.1	74.1	82	451	2.9	11.1	0.9	28.1	16.1	38	21
L1G8		1.10–1.25	1.48	1.02	0.084	0.1	0.05	0.175	1.26	0.013	7	2.7	7	1.5	183	48.2	34.2	39	232	0.5	2.3	0.2	10.8	7.2	13	2
L1G9		1.25–1.40	1.56	1.17	0.077	0.09	0.06	0.188	1.18	0.019	8	3.7	8	2.0	183	42.0	36.4	37	222	0.6	3.2	0.3	14.8	7.7	15	3
S01	Ilumetsa	0.25–0.40	2.28	1.54	0.147	0.15	0.11	0.269	1.86	0.025	15	5	12	2.6	248	97.1	51.6	50	324	1.0	4.7	0.3	19.8	8.9	30	2
S02	Small	0.40–0.55	2.19	1.15	0.137	0.14	0.09	0.287	1.83	0.025	10	5.5	11	2.1	217	76.4	52	48	299	0.9	4.6	0.2	13.0	9.8	19	2
S03		0.55–0.70	2.23	1.16	0.137	0.14	0.11	0.295	1.85	0.022	10	5.0	12	2.4	228	74.7	49.9	52	317	0.9	4.3	0.2	10.0	8.9	14	1
S04		0.70–0.85	2.40	1.43	0.139	0.14	0.15	0.263	1.95	0.016	13	6.2	15	2.8	255	84.2	54.6	48	313	0.9	5.0	0.3	12.5	9.1	18	2
S05		0.85–1.00	2.18	1.15	0.131	0.09	0.14	0.139	2.15	0.015	12	5.3	13	3.0	271	63.5	48.1	42	295	0.8	4.1	0.2	18.7	8.5	20	1
S06		1.00–1.15	2.12	1.28	0.116	0.11	0.10	0.249	1.86	0.017	12	4.5	10	2.1	278	73.7	47.2	45	300	0.8	3.8	0.3	12.8	8.4	15	1
S07		1.15–1.30	2.65	1.33	0.131	0.23	0.14	0.423	2.25	0.021	10	5.1	15	2.7	327	73.9	60.9	61	334	1.0	5.7	0.3	9.7	10.5	16	2
S08		1.30–1.40	2.34	1.06	0.141	0.14	0.14	0.272	1.85	0.022	11	4.5	14	3.1	268	85.6	50.1	48	293	0.9	5.6	0.2	9.4	8.8	14	2
S09[b]		1.40–1.50	3.52	2.16	0.245	0.16	0.4	0.239	2.89	0.032	25	13.3	34	6.4	334	109.4	76.6	55	373	1.4	7.3	0.3	22.5	10.7	35	3
S10[c]		0.80–0.85	3.07	1.91	0.176	0.14	0.32	0.185	2.34	0.034	21	11.1	33	5.1	361	92.7	66.5	46	298	1.5	8.2	0.5	15.1	10.8	26	4
S11[d]		0.90–0.95	2.52	1.14	0.153	0.16	0.15	0.309	2.23	0.016	11	5.8	14	2.4	171	90.5	55	54	332	1.0	4.1	0.2	28.4	9.9	30	2

[a]Sample with charcoal.
[b]Devonian siltstone fine-grained sandstone.
[c]Lens of reddish clayey till.
[d]Lens of gray clayey silt with charcoal.

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

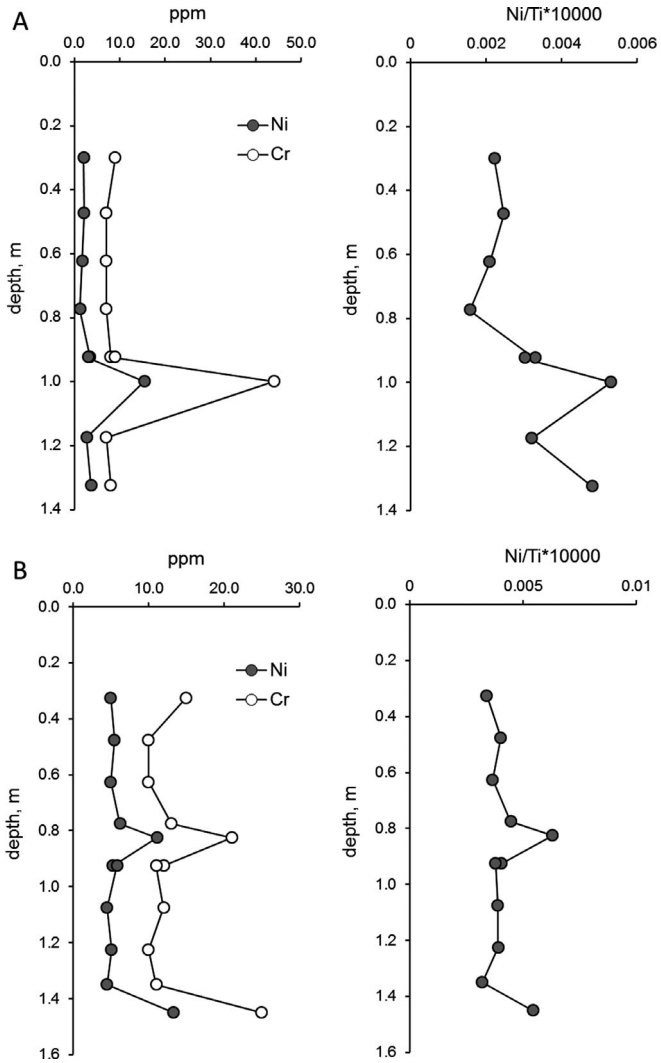


Fig. 8. Variation of Cr and Ni content and Ni/Ti ratio (expressed as Ni in ppm ratio to Ti*10000 in wt%) in (a) Ilumetsa Large and (b) Ilumetsa Small profiles. Shaded interval shows zone where most of the charcoal pieces were found.

unpublished study of IS by E. Versh, U. Preeden, and J. Plado). The absence of meteorites associated with the Ilumetsa structures can be explained by a relatively old age of the structures combined with high rates of weathering in the humid, well-drained conditions of SE Estonia if we are to consider them of impact origin. Additionally, the idea that the craters were produced by an impact of a stony asteroid (e.g., of chondritic composition) could be a possibility. Commonly, small craters on Earth (<300 m in final size) are formed by iron impactors that have a greater strength compared to that of stony bodies (that tend to be disrupted and decelerated during their passage through the atmosphere; Bland and Artemieva 2006). However, there are small craters known to be formed by stony and/or stony-iron impactors such as the 14 m in diameter Carancas in Peru made in 2007 by a H4-5 ordinary chondrite (Kenkmann et al. 2009), the 15 m Haviland in the United States made by an impact of pallasite, and a 24 m Dalgaranga West Australia associated with a mesosiderite (Bevan and Griffin 1994).

Meteorites, especially those containing metals, tend to weather rapidly in the terrestrial environment especially if liquid water is available (e.g., Lee et al. 2006). The residence ages of meteorite finds from the middle latitudes have not been studied in detail, because most research is focused on more significant meteorite collection sites such as hot and cold deserts (e.g., Jull 2006). Ordinary chondrites recovered from hot deserts have ~80% of their Fe oxidized within the first 5000 years of residence on the surface of the Earth

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. ^{14}C ages and calibrated ages of charcoal samples from both Ilumetsa structures.

Sample name	Lab.-Nr.	Crater & trench	Location within trench[a]	$d^{13}C^{b,c}$ (‰)	^{14}C-age[b] (BP)	Calibrated age BP[d]
Ilume_L_1_7[e]	VERA-6313	Ilumetsa large	2.5 m/65 cm	−24.9 ± 0.6	6147 ± 30	7160–6950
Ilume_L_1_13	VERA-6314		2.3 m/63 cm	−25.1 ± 0.6	6552 ± 28	7510–7420
Ilume_L_1_21	VERA-6315		2.2 m/65 cm	−27.9 ± 0.6	6739 ± 35	7670–7560
Ilume_L_1_33	VERA-6316		2.1 m/60 cm	−26.5 ± 1.0	6366 ± 30	7420–7250
Ilume_S_2_3 [e]	VERA-6325	Ilumetsa Small, shorter trench	3.5 m/55 cm	−24.9 ± 0.8	6212 ± 37	7250–7000
Ilume_S_1_6	VERA-6420	Ilumetsa Small, longer trench	5.0 m/80 cm	−25.8 ± 0.9	1643 ± 38	1620–1410
Ilume_S_1_11g	VERA-6317		5.6 m/78 cm	−25.6 ± 0.6	7663 ± 33	8540–8400
Ilume_S_1_17	VERA-6421		5.9 m/57 cm	−27.2 ± 0.8	7541 ± 39	8420–8210
Ilume_S_1_30_2	VERA-6366		5.6 m/75 cm	−21.7 ± 1.2	7246 ± 38	8170–7980
Ilume_S_1_32	VERA-6422		5.6 m/90 cm	−23.1 ± 0.6	6320 ± 40	7330–7160
Ilume_S_1_36	VERA-6423		6.3 m/85 cm	−25.2 ± 0.6	6329 ± 39	7410–7160
Ilume_S_1_43	VERA-6424		6.0 m/115 cm	−23.6 ± 0.8	5419 ± 37	6300–6120
Ilume_S_1_48	VERA-6324		8.0 m/50 cm	−25.8 ± 1.2	4161 ± 33	4830–4580

Calibration is based on IntCal13 atmospheric curve (Reimer et al. 2013), plot is from OxCal v4.3.2 (Ramsey and Lee 2013).

[a]The first number (m) is the x position within trench (counting from the end of trench farthest from the crater rim), second number is the depth below local surface.

[b]1σ-uncertainty.

[c]Determined with the AMS system.

[d]Calculated with OxCal4.2 and the INTCAL13 calibration curve, data correspond to the 2σ-confidence level (95.4%), probability of the individual time periods in brackets, results rounded to 10 years.

[e]Samples (one most representative sample from every structure—see Discussion section) used to define age of the Ilumetsa crater features.

(fig. 2 in Bland et al. 2000, 2006). The oxidation rate of iron-bearing meteorites is even higher in a humid climate like Estonia, and it is therefore likely that all or nearly all of the metal would be oxidized since the structures' formation about 7000 years ago. The acidic, bog environment of the site would be particularly detrimental to this kind of meteorite. Additionally, the speed of oxidation process will be further enhanced by soil microorganisms colonizing the meteorite (Benzerara et al. 2006; Gronstal et al. 2009). All these factors would significantly limit the ability of a metal detector to pick up a signal of any potential meteorite.

Moreover, finding meteorites is especially problematic if an impactor was significantly disrupted (ablated and fragmented) during its atmospheric passage and during crater formation. This is an important factor for iron asteroids, but it is crucial in the case of stony bodies, which are considerably more fragile (Artemieva and Shuvalov 2001; Bland and Artemieva 2006). As a result, stony meteorites tend to reach the surface as vapors mixed with small meteorites (Artemieva and Shuvalov 2019) that were significantly decelerated, ablated, and fragmented below the limit required to produce an explosion crater. A recent example of such an event was the explosion of a 20 m diameter asteroid above Chelyabinsk in Russia (Popova et al. 2013).

The division of one larger body into numerous smaller ones has important implications for the ability of identifying meteorites around the crater. Firstly, it would increase the weathering rate due to the increased surface–volume ratio of the divided pieces. This shortens the time for which extraterrestrial particles are distinguishable from other terrestrial rocks at the hand sample scale. Secondly, it limits the percentage of detectable meteoritic pieces to only those that are bigger than a certain threshold (depending on type of instrument used). This factor is especially important when searching for ordinary chondrites, which need to be of a sufficient size (and/or to be located at very shallow depths) to be sensed by a detector due to their low metal content.

The size–frequency distribution of crater-related meteorites can vary significantly even for crater sizes similar to those at Ilumetsa. For example, both the Morasko and Kaali structures are comparable strewn fields, with the largest crater ~100 m in diameter, where both were formed by an IAB-MG iron impactor (Spencer 1938; Karwowski et al. 2016); however, the number of available large pieces varies significantly. There are multiple pieces found at Morasko that are hundreds of kilograms in weight (total mass according to the Meteoritical Bulletin Database is 290 kg, but a true value, including a single piece of 261 kg found in 2012 and 271 kg found in 2017 is much higher), while the largest piece of the impactor found in Kaali so far is only 600 g. That fragment was found in 2017 during our

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

own fieldwork campaign there by M. Szyszka. Before 2017, the largest known fragment from this location was of ~20 g total mass, and according to the Meteoritical Bulletin Database, the total mass of fragments was 2.5 kg (but likely was lower than that); including our recent finds it is now close to ~4 kg in mass.

As a result, the number of small impact craters associated with stony impactors is certainly underestimated. This is because the presence of meteorites associated with small craters provides the key confirmation of their impact origin, where as discussed above, iron meteorites are much easier to find than stony meteorites (especially if the latter ones are significantly disintegrated). An example of this kind of situation is the Carancas event in 2007 in Peru (Tancredi et al. 2008; Kenkmann et al. 2009). This was quite an unusual case as the impact occurred on land at an elevation of 3000 m a.s.l. Here, an H4–5 chondrite impactor, a couple of meters in diameter, formed a single, 14 m diameter crater. No magnetic anomaly was found within the crater implying that there is no large meteorite fragment below the crater floor (Rosales et al. 2008). However, numerous, small meteorite fragments were found distributed in the ejecta and around the structure, with powdered meteoritic material found covering crater walls (Tancredi et al. 2008). Numerous pieces of meteorite (including couple of pieces up to ~350 g), weighing several kilograms in total, have been discovered from the site (Kenkmann et al. 2009). However, photographs of the Carancas meteorite (available on the Meteoritical Bulletin Database) taken only 1–2 years after the meteorite fall show signs of iron oxidation (although the date that the pieces were collected and the duration of their exposure to moisture within the environment is unknown). Therefore, it is very probable that if the formation of Carancas crater had not been witnessed, it would have been hard or impossible to recognize the associated meteorites, especially after they had been subject to weathering at the surface for a couple of thousands of years in humid conditions. As such, we suggest that if the Ilumetsa structures were made by a chondritic impactor, that fragmented significantly during its passage through the atmosphere and during the impact, and later weathered in a humid environment for ~7000 years, then the chances of finding any coherent, recognizable meteorite fragments are probably very low.

This may explain why we were not able to identify any geochemical signatures of the meteoritic component, in the ejecta blanket deposits at both studied sites. We found normal sedimentary background levels of Ni, an element that is typically enriched in iron-bearing meteorites (irons, stony-irons, and large percentages of chondrites; e.g., Jarosewich 1990). Moreover, we were unable to identify any enrichment in the PGEs at IL where Ir concentrations remained at the average abundance for that of continental crust (in the order of 0.03–0.05 ppb; Taylor and Mclennan 1985; Wedepohl 1995). This contrasts to meteoritic materials that are enriched in PGEs at a level of at least two-order of magnitude over the Earth's background level (Palme 2008). It could be possible that the impactor was of non-chondritic (eucritic) composition, which when compared to other meteorites types, are characterized by low siderophile element (including PGE) contents and Ir concentrations that are often as low as that of upper continental crust (0.03 ppb; Muñoz-Espadas et al. 2003). If this is the case, then any meteoritic matter would be dispersed and diluted in the target rocks to a level that would be unrecognizable by means of bulk geochemical analysis. Additionally, it would be highly likely that a nonuniform distribution of extraterrestrial material would be expected around the small impact craters.

An alternative explanation could be that any PGEs have been redistributed by secondary processes (e.g., Racki et al. 2011). For example, diagenetic remobilization and secondary enrichment of Ir in sedimentary environments are a widely discussed phenomenon (e.g., Wallace et al. 1990; Kramar et al. 2001). Indeed, in the sedimentary cycle, Ir is precipitated preferentially with iron oxides and ferromanganese phases while sediments rich in organic matter and sulfides can be also enriched in Ir owing to redox-controlled remobilization and precipitation (Colodner et al. 1992). In this case, a bog environment rich in different organic substrates and with typically low pH/Eh conditions, or the acidic podzols developed below coniferous forests could have caused the mobilization of PGEs during the weathering of the meteoritic material. In such a case, we might expect preferential uptake in secondary phases away from the impact event horizon. Despite this potential scenario, we were not able to locate any enrichments in PGEs throughout the studied section or within the undisturbed pre-impact sediments.

Gurdziel and Karwowski (2011) have shown in the Morasko meteorite fragments that the minerals most resistant to weathering are phosphides (schreibersite, rhabdite, nickelphosphide) and carbides (cohenite); where the phosphides are replaced in the course of weathering to a metallic phosphorus-free Ni phase which is similar to native nickel or avaurite. This suggests that the few Ni-rich microscopic particles found at the IL structure could be remnants of Ni-phosphides. However, further research and more microscopic fragments would be needed to be found and analyzed in order to prove their origin.

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Comparison of Charcoals in Kaali Main, Kaali 2/8, Morasko Main, and Ilumetsa Structures

We have studied the proximal ejecta blankets of the Kaali and Morasko structures which have been confirmed as small impact craters (Muszynski et al. 2012; Plado 2012). Within these eject a blankets, we found charcoals in the same geomorphological and stratigraphical setting in relation to the respective crater size and rim location. Additionally, the charcoals found at the Kaali Main crater (Losiak et al. 2016), the Kaali 2/8 double crater (unpublished data based on 2016 fieldwork), and at the Morasko crater (Szokaluk et al. [2015, 2019] and field visits by AL) have been found to have ages consistent with independently determined ages of the formation of these craters. Despite the lack of obvious meteroritic evidence found in the region of the Ilumetsa structures, we show that charcoal is similarly distributed within proposed ejecta blanket of both Ilumetsa structures.

In all cases, charcoals were found only within a specific zone within the strata ejected during the impact and were characteristically absent outside of this zone farther away within the ejecta blanket (e.g., Kaali Main, Losiak et al. 2016); within both Ilumetsa structures, charcoal was notably absent within the first few meters of the excavated trenches (the farthest away from the crater rim). The vertical distribution of the particles is such that charcoals are the most numerous in a 10–20 cm thick layer near the base of the ejecta blanket, with fewer pieces found at shallower depths. The lateral distribution of charcoals in the ejecta blankets is more complex. In all studied cases, charcoal pieces started to appear when thickness of the ejecta blanket exceeded 50 cm, which potentially held enough banked heat to allow for the pyrolysis of organic material. Charcoal pieces were not apparent at shallower depths, but it could be that they were not easily detectable (e.g., due to overprinting by the soil-forming processes), or it could be that they were formed but due to greater oxygen availability at shallower depths within the ejecta blanket that this allowed complete combustion of the organic material, instead of preservation as charcoal, which is formed only by pyrolysis in the absence of oxygen.

Despite the similarity of the location of the charcoal-containing horizons, the extent of the charcoal-containing sediment layer in relation to the distance from the crater structure's rim varies. Within Kaali Main, pieces of charcoal are present within a ~3 m long section of a ~6 m long trench (Losiak et al. 2016) and within another ~3 m long section that later extended the previous trench in the direction of the rim (unpublished data, 2017). Although the number of particles varies significantly on the cm scale a similar situation is present in the Kaali 2/8 double crater (unpublished data based on fieldwork from 2016). Charcoal pieces found in Morasko were too rare to determine the size of the charcoal-bearing layer, but pieces that were found were present in a zone starting from crater rim up to ~10 to 20 m from the rim.

In the IL structure, we identified a layer with charcoal pieces that were mostly ~1 mm in diameter dispersed every few cm from each other (Fig. 3), in a lens/layer similar in thickness and preserving a charcoal density similar to that found within the Kaali Main till-rich ejecta horizon. However, in the Ilumesta structure, on both sides of the ~80 cm wide trench, the macroscopically detectable charcoal-rich horizon ends ~70 cm from the rim edge of the outcrop, while in Kaali and Morasko, this layer continued until the end of the excavated ejecta sequence. The situation is similar in the shorter trench excavated in the Ilumetsa Small structure; although instead of numerous small pieces of charcoal, we found only a few larger pieces. In the longer trench at Ilumetsa Small, the situation appears more complex. It was clear in the field that charcoals belong to potentially several phases of formation. Particularly notable was a large zone of grayish soil containing numerous charcoal particles, including pieces a couple of centimeters in length. However, the presence of the charcoals at Ilumetsa, in a similar geomorphological location to those found close to the crater rim at Kaali and Morasko, suggests that Ilumetsa charcoals were likely formed by the same process as those found in the Kaali Main, Kaali 2/8, and the Morasko Main structures. As yet the mechanism of formation of these charcoals has not yet been explained in regard to cratering processes, beyond that previously discussed in Losiak et al. (2016).

Age of the Ilumetsa Structures

The radiocarbon dates from the Ilumetsa charcoals suggest that both crater-like structures were formed simultaneously (significant within two sigma) between 7170 and 7000 cal BP. This age is calculated by combining the ages of charcoal from the two different structures (located 750 m from each other). The Ilume_L_1_7 and Ilume_S_2_3 (marked with two red rectangles on Fig. 9) samples were selected as the most representative for the age of the structures because they (1) belong to a likely "impact-charcoal zone" that is geomorphologically similar to that identified at the Kaali Main, Kaali 2/8, and Morasko craters; (2) are the youngest of the dated charcoals that, at the same time; (3) are slightly older than the ^{14}C ages of gyttja (7170–6660 cal. years BP, 6030 ± 100 ^{14}C age) obtained from

inside the larger Ilumetsa structures (Liiva et al. 1979) that by its nature most postdate the formation of the impact crater. The radiocarbon ages of these two samples overlap within the dating uncertainty: 7160–6950 cal. years BP or 6147 ± 30 ^{14}C age (Ilume_L_1_7) and 7250–7000 cal. years BP or 6212 ± 37 ^{14}C age (Ilume_S_2_3). The fact that gytta deposition at Ilumetsa started very soon after the formation of the crater feature is similar to that also found in the Kaali Main case (Losiak et al. 2016), suggesting further similarities between these cases.

The charred wood of the Ilume_L_1_7 and Ilume_S_2_3 samples suggests that they come from the youngest trees, or the youngest parts of older trees growing in the area when the Ilumetsa structures were formed. Five other samples (Ilume_L_1_13, Ilume_L_1_21, Ilume_L_1_33, Ilume_S_1_32, Ilume_S_1_36) are up to couple of 100 years older than the determined craters' age. This is consistent with charcoal formed from older trees during the formation of Ilumetsa. It is worth noting that all of the dated charcoals from the IL "impact-charcoal zone" yielded the same ^{14}C age (when adjusted for measurement uncertainty and the "old wood problem"). The ^{14}C ages distribution of the IL is similar to the one from Kaali Main (Losiak et al. 2016). We interpret these seven charcoal samples as being created during the formation of the structures at Ilumetsa structures and are likely sourced from trees that were killed during this event. The timing of charring in respect to an impact scenario is not yet known and will require additional research and is part of ongoing debates regarding the energy regime of small impact cratering processes (Losiak et al. 2016)

The remaining radiocarbon - dated samples (Ilume_S_1_11g, Ilume_S_1_17, Ilume_S_1_30_2 as well as Ilume_S_1_6, Ilume_S_1_48, Ilume_S_1_43), despite being found in the longer trench at IS, were certainly not charcoalized during the formation of Ilumetsa. Three of these samples (Ilume_S_1_11g, Ilume_S_1_17, Ilume_S_1_30_2) from IS are ~1000 years older than the crater age as identified in this study. Moreover, all of them were found close to each other within a distinct, grayish, organic, and charcoal-rich lens (Fig. 4, Ilume_S_1_11g). This zone within the longer IS trench was visibly different from the other regions of the trench where the charcoal pieces were found. Numerous charcoals from this location were particularly large—up to ~2.5 cm in length and ~1 cm in diameter and contained identifiable partial tree rings visible with the naked eye. The curvature of the rings seems to suggest that they come from larger trees (thicker branches or pieces of small trunks) than those elsewhere in the ejecta sequence. We interpret these samples as parts of the trees that died some time (<1000 years) before the formation of the structures at Ilumetsa most likely by a forest fire. Charcoal is stable in the soil and in most cases is well preserved over thousands of years (De Lafontaine and Asselin 2011) and in sediments over 100s of millions of years (Scott 2000).

Three samples from the Ilumetsa Small structure are younger than the age provided by Liiva et al. (1979). Ilume_S_1_6 is the youngest of them, dated at 1620–1410 cal. years BP (1643 ± 38 ^{14}C age), which is more than 5000 years younger than the determined age of the craters. This sample was collected farther away from the structure's rim than the expected location of the "charcoal zone" based on the comparison with the Kaali Main crater and Morasko structures. These ages were dated from two 2–3 mm long pieces of charcoal that were found within a grayish sediment lens and we interpret these to be formed during a forest fire well after the formation of the crater structures. These must the have been moved downward deeper below the surface by bioturbation (Zazula et al. 2007; Haussmann 2016). The Ilume_S_1_48 charcoal sample produced an age of 4830–4580 cal. years BP (4161 ± 33 ^{14}C age), more than 2000 years after the interpreted age of the craters. This was found at the end of the trench located closest to the rim, and at the depth of 50 cm, just below the A horizon of the soil (which is very thick in this location). This sample was selected in order to test whether these charcoals have different age properties than those charcoals found in the expected location at the bottom of the ejecta blanket. The age of this sample also suggests that its genesis was from natural forest fires that were later incorporated into the soil. The reason for the relatively young age of the third sample Ilume_S_1_43 of 6300–6120 cal. years BP (5419 ± 37 ^{14}C age) is less clear. One suggestion is that this sample is due to potential accidental contamination in the field as almost all of our dated samples were a combination of multiple charcoal pieces usually found within a couple of mm from each other. All charcoals described as one sample are taken from a single location (less than a couple of cm^3). They, while in the outcrop, formed a coherent piece, or a closely adhering group of small particles (like ones in Figs. 3 and 4). Charcoal samples were taken in the field together with a matching sample of the surrounding sediment. The charcoal samples tend to be very fragile and after their extraction and drying in the laboratory, some of them divide into smaller pieces. For example, an addition of only a small particle of a recent charcoal from the surface (or other organic carbon contaminant) could significantly skew the obtained age (all of the dated samples consisted of at least a couple of small particles, so intermixing of a small, young piece during sampling cannot be excluded).

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

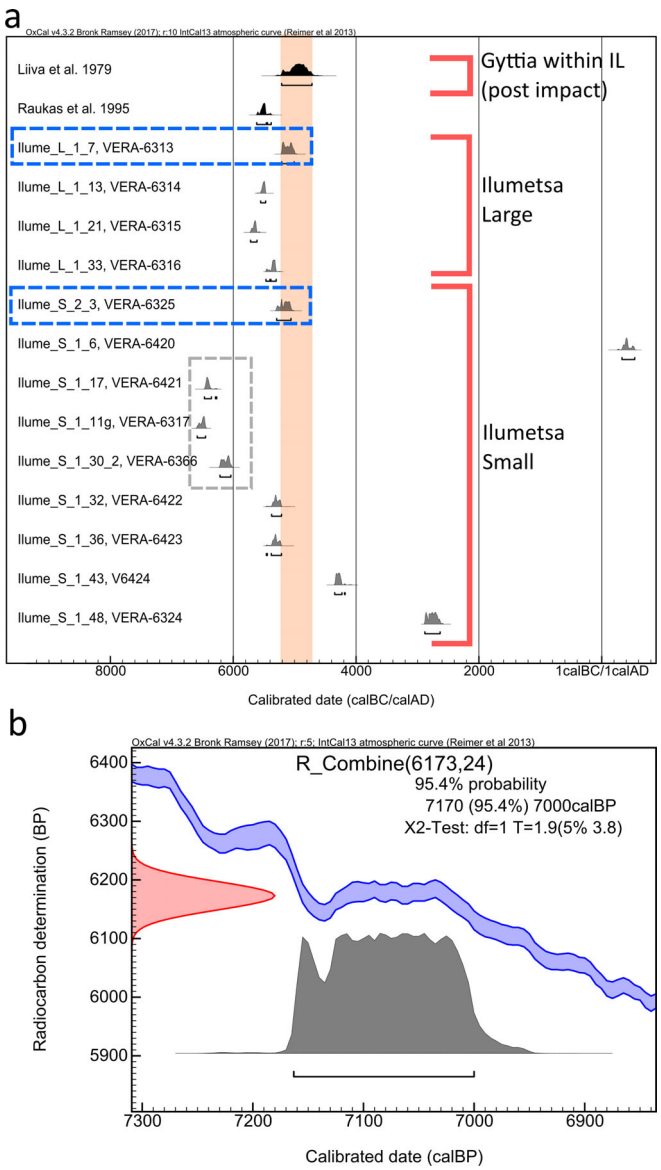


Fig. 9. A) Calibrated ages of charcoal found within proximal ejecta of both Ilumetsa structures. Black probability distributions show age estimates made by previous researchers: Liiva et al. (1979); gyttja and peat from the very bottom of IL and Raukas et al. (2001); glassy spherules found in the Meenikunno bog, ~6 km SWW from the structures, whose genetic relation with Ilumetsa structures has not been unequivocally established. The orange filled rectangle corresponds to the Liiva et al. (1979) age which is postdating the impact. Light gray probability distributions represent samples measured in this study: four top samples come from Ilumetsa Large, one sample comes from shorter trench within Ilumetsa Small, eight samples come from longer trench within the same crater. Samples enclosed within gray rectangle were found within a gray zone highly enriched in charcoal with multiple, well-preserved pieces more than 2 cm in length. Three samples yielded ages younger than the age of gyttja developed within the large Ilumetsa structure. Please note that the full laboratory name of the sample (e.g., VERA-6313 as given in Table 2) was shortened to V6313 for the sake of clarity. B) A terminus post quem age (7170–7000 cal BP; mean value: 6173 ± 24 ^{14}C yr BP, chi-squared test passed) of the Ilumetsa craters is based on the combined results of the samples representing Ilumetsa Large (Ilume_L_1_7) and Ilumetsa Small (Ilume_S_2_3) structures (marked on the [a] with blue-dotted rectangles). Those samples were selected because they are (1) older than the oldest sediment from inside the Ilumetsa Large (Liiva et al. 1979) so they predate formation of this structure; and (2) they are youngest of the samples oldest than Liiva et al. (1979), so the time lag related to the old wood effect is minimized. Calibration is based on IntCal13 atmospheric curve (Reimer et al. 2013), plot is from OxCal v4.2.4 (Ramsey and Lee 2013). (Color figure can be viewed at wileyonlinelibrary.com.)

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The age of the Ilumetsa structures determined here is in agreement with Liiva et al. (1979: 7170–6660 cal. years BP), but ~500 years younger than the age (6542 ± 50 ^{14}C years or 7570–7320 cal. years BP; IntCal13) proposed by Raukas et al. (2001). Raukas et al. (2001) dated peat from the Meenikunno bog, ~4.5 km away from the craters, where they found multicolored porcelain-, opaline-, and marble-like glassy spherules up to some millimeters in size that were ellipsoidal, and drop-like or rounded in shape at the depth of 5.7 m. The chemical composition of spherules was not reported but was merely described as being controlled by the mineralogy of the target rocks, consisting mainly of silica and calcium with an admixture of Fe and Ni and several other elements. Thus, based solely on morphology, the spherules were interpreted as impact-related dissipated melt or condensed vapor, without discussing other possible origins. At present, given the age difference between the spherule-bearing peat and the structures at Ilumetsa, and without any analytical proof of the spherules' impact origin, we suggest that the Meenikunno spherules should not be considered as related to the Ilumetsa event.

Are the Ilumetsa Structures of Impact Origin?

The identification of small impact craters formed in unconsolidated materials is a challenging task because standard impact cratering indicators that are established from analysis of larger structures formed in consolidated materials are not helpful. There are no shatter cones, the amount of shocked quartz is small and spread out over a large area, and the heating from the impact is suggested to be minimal. For example, in the Morasko strewn field (Poland), the volume of sediment shocked >5 GPa (required for producing PDFs—in quartz) is negligible (Bronikowska et al. 2017) and as a result, a hunt for shocked grains and impact melt was unsuccessful (Muszynski, personal communication). Therefore, the lack of PDFs or other clear indications of shock metamorphism at the Ilumetsa site is to be expected.

Meteorite fragments found associated with small impact craters formed in unconsolidated sediments, therefore, tend to be the only criteria applied to their identification. The lack of observable extraterrestrial materials at the Ilumetsa site is clearly the largest issue when it comes to proving the impact origin of the structures. However, we would argue that this lack cannot be used as evidence that Ilumetsa is definitely not an impact crater. This is because the search for meteorites near impact craters is problematic for two reasons (i) the meteorite survival period and (ii) the difficulty of finding fragments. First, the meteorite survival period is expected to be shortened in the Ilumetsa structures' case, as they were emplaced within a water-soaked bog in Estonia that has undergone multiple freeze–thaw cycles every year. The Kaali and Morasko strewn fields that are relatively close by are examples of two recent, small impact craters that like Ilumetsa were developed in unconsolidated materials. However, these are not only much younger than Ilumetsa but also were formed in relatively dry, well-drained areas making the preservation potential of meteorites or meteorite fragments much higher. Even better meteorite preservation potential is associated with craters made in an arid or semiarid climate, such as Dalgaranga and Haviland. Second certain meteorite types are easier to find than others. Large pieces of iron meteorites are the easiest to locate (with metal detectors) and can be found even if they are buried under a couple of meters of soil. Small stony meteorites are much harder and sometimes even impossible to find unless they are positioned directly on the surface. As such we suggest that the lack of identifiable meteorite fragments at Ilumetsa is insufficient to argue against the structures' being from an impact origin.

It is, however, clear that while the evidence for an impact origin for the Ilumetsa structures remains circumstantial, there are features that support this theory. First, the geomorphologic properties of both craters, in a form of a rimmed depression with a specific diameter/depth ratio (Holsapple et al. 2002), are consistent with them being formed by an impact. Where the fact that the rim of the structures is made of a mixture of Quaternary sands and tills containing numerous clast-like blocks of Devonian sandstones and clays, and where the glacial deposits within the crater are reduced in thickness supports an explosive origin of the Ilumetsa structures (Aaloe 1963). Second, the existence of a charcoal-bearing layer within the interpreted proximal ejecta of both Ilumetsa craters that are similar to charcoal-bearing layers found in other craters that have proven impact origin (Kaali Main, Kaali 2/8, Morasko Main) is intriguing. This charcoal-bearing layer is also found at the same geomorphological location, implying that a similar process likely formed the charcoals and structures that contain them.

Other possible mechanisms that might explain the formation of structures at Ilumetsa seem impossible or highly improbable. There are, for example, numerous depressions in the landscape of northern Europe that were made by glacial or periglacial processes, and similarly sized holes may be formed as pingos or glacial kettle holes. However, in the case of the Ilumetsa structures, such mechanisms can be excluded as we have established the ages of the structures as being ~7 ka, and

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

this is ~7 ka after the deglaciation of this area (Stroeven et al. 2016). Moreover, a volcanic or karst origin of the structures is impossible because (i) there has been no geologically recent volcanic activity in Estonia and (ii) the Ilumetsa site does not contain rocks that are karst susceptible where we note that the crater forms cannot be sinkholes because sinkholes do not have rims. It seems unlikely that the two crater-like structures were dug ~7 ka ago by human activity. However, the stratigraphy of the proposed ejecta would appear at odds with such a hypothesis and people would have to achieve this without leaving any artifacts behind that can be data from this time period. Citing Sherlock Holmes, "when you have eliminated the impossible, whatever remains, however improbable, (is probably) the truth." Of course, even citing Sherlock Holmes is not enough to unequivocally prove that Ilumetsa structures are impact craters, but we believe that the evidence is sufficient to call them "rather likely" impact craters.

CONCLUSIONS

A GPR survey of the Süvahavva bog indicates that there are no additional crater-like structures that are larger than 40 m in diameter at Ilumetsa. However, despite a rather dense GPR profile spacing, we cannot exclude the existence of a peat-buried crater of <40 m in diameter. We detected no clear signal of extraterrestrial material within the proximal ejecta of the Ilumetsa structures, from chemical analyses or a search for meteoritic fragments. We suggest that this may be because the structures at Ilumetsa were formed by an impact of a stony-iron or stony body, which has since been significantly weathered in the wet-temperate climate of Estonia.

Using charcoals found within the sandy material that we interpret as a proximal ejecta sequence, we have dated the Ilumetsa structures as being postglacial in age; therefore, they cannot be of glacial origin. Charcoal from the two structures which are ~750 m apart from one another is of the same age (between 7170 and 7000 cal BP), providing clear evidence that they share a common origin. The age is similar to that (7170–6660 cal. years BP) obtained from the deepest organic sediments in the large structure at IL (Liiva et al. 1979).

Acknowledgments—Support for this study was provided by the 2016 Barringer Family Fund for Meteorite Impact Research as well as project "Kaali" founded by the Institute of Geological Sciences Polish Academy of Sciences. A.L. acknowledges funding from the European Union's Horizon 2020 research and innovation program under the Marie Sklodowska-Curie grant ImpChar, agreement no. 749157. Work by A.J. and J.P. is supported by the Institutional Research Funding Program (IUT20-34). We thank Tartu University for use of the analytical instrumentation in the Estonian Center of Analytical Chemistry. We thank Stefan Lehr for his help in the ^{14}C sample preparation. Finally, we are most grateful to the Devils that are believed to use the Ilumetsa structures as an entrance to hell, for not interfering with our fieldwork.

Editorial Handling—Dr. A. J. Timothy Jull

REFERENCES

Impact Earth database

Aaloe A. 1963. New data on the structure of Ilumetsa craters. *Eesti NSV Teaduste Akadeemia Geoloogia Instituudi uurimused* 11:35–43 (in Russian).

Artemieva N. A. and Shuvalov V. V. 2001. Motion of a fragmented meteoroid through planetary atmosphere. *Journal of Geophysical Research* 106:3297–3309.

Artemieva N. and Shuvalov V. 2019. Atmospheric shock waves after impacts of cosmic bodies up to 1000 m in diameter. *Meteoritics & Planetary Science* 54:592–608.

Benzerara K., Chapon V., Moreira D., López-Garcia P., Guyot F., and Heulin T. 2006. Microbial diversity on the Tatahouine meteorite. *Meteoritics & Planetary Science* 41:1249–1265.

Bevan A. and Griffin B. 1994. Re-examination of the Murchison Downs meteorite: A fragment of the Dalgaranga mesosiderite. *Journal of the Royal Society of Western Australia* 77:45–49.

Bland P. A. and Artemieva N. A. 2006. The rate of small impacts on Earth. *Meteoritics & Planetary Science* 41:607–631.

Bland P. A., Bevan A. W. R., and Jull A. J. T. 2000. Ancient meteorite finds and the Earth's surface environment. *Quaternary Research* 53:131–142. https://doi.org/10.1006/qres.1999.2106.

Bland P. A., Zolensky M. E., Benedix G. K., and Sephton M. A. 2006. Weathering of chondritic meteorites. In *Meteorites and the early solar system II*, edited by Lauretta D. S. and McSween H. Y. Jr. Tucson, Arizona: University of Arizona Press. pp. 853–867.

Bronikowska M., Artemieva N. A., and Wünnemann K. 2017. Reconstruction of the Morasko meteoroid impact—Insight from numerical modeling. *Meteoritics & Planetary Science* 52:1704–1721.

Colodner D. C., Boyle E. A., Edmond J. M., and Thomson J. 1992. Post-depositional mobility of platinum, iridium, and rhenium in marine sediments. *Nature* 358:402–404.

De Lafontaine G. and Asselin H. 2011. Soil charcoal stability over the Holocene across boreal northeastern North America. *Quaternary Research* 76:196–200.

Gronstal A., Pearson V., Kappler A., Dooris C., Anand M., Poitrasson F., Kee T. P., and Cockell C. S. 2009. Laboratory experiments on the weathering of iron meteorites and carbonaceous chondrites by iron-oxidizing bacteria. *Meteoritics & Planetary Science* 44:233–247.

Gurdziel A. and Karwowski L. 2011. The secondary changes of phosphides and carbides in Morasko meteorite. *Acta Societatis Metheoriticae Polonorum* 2:25–33 (in Polish).

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Habicht H.-L., Rosentau A., Jõeleht A., Heinsalu A., Kriiska A., Kohv M., Hang T., and Aunap R. 2017. GIS-based multiproxy coastline reconstruction of the eastern Gulf of Riga, Baltic Sea, during the Stone Age. *Boreas* 46:83–99.

Haussmann N. S. 2016. Soil movement by burrowing mammals: A review comparing excavation size and rate to body mass of excavators. *Progress in Physical Geography: Earth and Environment* 41:29–45. https://doi.org/10.1177/0309133316662569.

Hiiemaa H., Mustasaar M., Kohv M., Hang T., Jõeleht A., Lasberg K., and Kalm V. 2014. Geological settings of the protected Selisoo mire (northeastern Estonia) threatened by oil shale mining. *Estonian Journal of Earth Sciences* 63:97–107.

Holsapple K., Giblin I., Housen K., Nakamura A., and Ryan E. 2002. Asteroid impacts: Laboratory experiments and scaling laws. In *Asteroids III*, edited by Bottke Jr. W. F., Cellino A., Paolicchi P., and Binzel R. P. Tucson, Arizona: The University of Arizona Press. pp. 443–462.

Jarosewich E. 1990. Chemical analyses of meteorites: A compilation of stony and iron meteorite analyses. *Meteoritics* 25:323–337. https://doi.org/10.1111/j.1945-5100.1990.tb00717.x.

Jull A. J. T. 2006. Terrestrial ages of meteorites. In *Meteorites and the early solar system II*, edited by Lauretta D. S. and McSween H. Y. Tucson, Arizona: University of Arizona Press. pp. 889–905.

Karwowski L., Kryza R., Muszyński A., Kusz J., Helios K., Drożdżewski P., and Galuskin E. V. 2016. Czochralskiite, $Na_4Ca_3Mg(PO_4)_4$, a second new mineral from the Morasko IAB-MG iron meteorite (Poland). *European Journal of Mineralogy* 28:969–977. https://doi.org/10.1127/ejm/2016/0028-2557.

Kenkmann T., Artiemieva N. A., Wunnemann K., Poelchau M. H., Elbashausen D., and Nunez del Prado H. 2009. The Carancas meteorite impact crater, Peru: Geologic surveying and modeling of crater formation and atmospheric passage. *Meteoritics & Planetary Science* 44:985–1000.

Kramar U., Stüben D., Berner Z., Stinnesbeck W., Phillipp H., and Keller G. 2001. Are Ir anomalies sufficient and unique indicators for cosmic events? *Planetary and Space Science* 49:831–837.

Lee M. R., Smith C. L., Gordon S. H., and Hodson M. E. 2006. Laboratory simulation of terrestrial meteorite weathering using the Bensour (LL6) ordinary chondrite. *Meteoritics & Planetary Science* 41:1123–1138.

Liiva A., Kessel H., and Aaloe A. 1979. Age of the Ilumetsa craters. *Eesti Loodus* 12:762–764 (in Estonian).

Losiak A., Wild E. M., Huber M. S., Wisniowski T., Paavel K., Jõeleht A., Välja R., Plado J., Kriiska A., Wilk J., Zanetti M., Geppert W. D., Kulkov A., Steier P., and Pirkovic I. 2016. Dating Kaali Crater (Estonia) based on Charcoal emplaced within proximal ejecta blanket. *Meteoritics & Planetary Science* 51:681–695. https://doi.org/10.1111/maps.12616.

Losiak A., Jõeleht A., Plado J., Szyszka M., Wild E. M., and Steier P. 2018. Dating small impact craters on Earth and the "old wood problem" (abstract #6219). 81st Annual Meeting of The Meteoritical Society 2018 (LPI Contribution no. 2067).

Melosh H. J. 1989. *Impact cratering. A geologic process*. Oxford: Clarendon Press. 245 p.

Muñoz-Espadas M.-J., Martínez-Frías J., and Lunar R. 2003. Main geochemical signatures related to meteoritic impacts in terrestrial rocks: A review. In *Impact markers in the stratigraphic record*, edited by Koeberl C. and Martínez-Ruiz F. C. Berlin: Springer. pp. 65–90.

Muszynski A., Kryza R., Karwowski Ł., Pilski A. S., and Muszynska J., eds. 2012. *Morasko. Najwiezkszy deszcz meteorytow zelaznych w Europie srodkowej (Morasko. The largest iron meteorite shower in central Europe)*. Poznan: Bogucki Wydawnictwo Naukowe. 111 pp.

Osinski G. R. and Grieve G. A. F. 2019. Impact Earth: A new resource for outreach, teaching, and research. *Elements* 15:70–71. https://doi.org/10.2138/gselements.15.1.70.

Palme H. 2008. Platinum-group elements in cosmochemistry. *Elements* 4:233–238.

Pike R. J. 1977. Size-dependence in the shape of fresh impact craters on the moon. Impact and explosion cratering: Planetary and terrestrial implications. In *Proceedings of the Symposium on Planetary Cratering Mechanics*. New York: Pergamon Press, Inc. pp. 489–509.

Plado J. 2012. Meteorite impact craters and possibly impact-related structures in Estonia. *Meteoritics & Planetary Science* 47:1590–1605.

Plado J., Sibul I., Mustasaar M., and Jõeleht A. 2011. Ground-penetrating radar study of the Rahivere peat bog, eastern Estonia. *Estonian Journal of Earth Sciences* 60:31–42.

Popova O. P., Jenniskens P., Emel'yanenko V., Kartashova A., Biryukov E., Khaibrakhmanov S., Shuvalov V., Rybnov Y., Dudorov A., Grokhovsky V. I., Badyukov D. D., Yin Q. Z., Gural P. S., Albers J., Granvik M., Evers L. G., Kuiper J., Kharlamov V., Solovyov A., Rusakov Y. S., Korotkiy S., Serdyuk I., Korochantsev A. V., Larionov M. Y., Glazachev D., Mayer A. E., Gisler G., Gladkovsky S. V., Wimpenny J., Sanborn M. E., Yamakawa A., Verosub K. L., Rowland D. J., Roeske S., Botto N. W., Friedrich J. M., Zolensky M. E., Le L., Ross D., Ziegler K., Nakamura T., Ahn I., Lee J. I., Zhou Q., Li X. H., Li Q. L., Liu Y., Tang G. Q., Hiroi T., Sears D., Weinstein I. A., Vokhmintsev A. S., Ishchenko A. V., Schmitt-Kopplin P., Hertkorn N., Nagao K., Haba M. K., Komatsu M., and Mikouchi T.; Chelyabinsk Airburst Consortium 2013. Chelyabinsk airburst, damage assessment, meteorite recovery, and characterization. *Science* 342:1069–1073.

Racki G., Machalski M., Koeberl C., and Harasimiuk M. 2011. The weathering–modified iridium record of a new Cretaceous-Palaeogene site at Lechówka near Chełm, SE Poland, and its palaeobiologic implications. *Acta Palaeontologica Polonica* 56:205–215.

Ramsey C. B. and Lee S. 2013. Recent and planned developments of the program OxCal. *Radiocarbon* 55:720–730.

Raukas A., and Teedumäe A. (Eds). 1997. *Geology and mineral resources of Estonia*. Tallinn: Estonian Academy Publishers. 436pp. ISBN 9985-50-185-3. sarv.gi.ee/geology.

Raukas A., Tiirmaa R., Kaup E., and Kimmel K. 2001. The age of the Ilumetsa meteorite craters in southeast Estonia. *Meteoritics & Planetary Science* 36:1507–1514.

Reimer P. J., Bard E., Bayliss A., Beck J. W., Blackwell P. G., Ramsey C. B., Grootes P. M., Guilderson T. P., Haflidason H., Hajdas I., Hatt C., Heaton T. J., Hoffmann D. L., Hogg A. G., Hughen K. A., Kaiser K. F., Kromer B., Manning S. W., Niu M., Reimer R. W., Richards D. A., Scott E. M., Southon J. R., Staff R. A.,

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Turney C. S. M., and van der Plicht J. 2013. IntCal13 and Marine13 radiocarbon age calibration curves 0-50,000 years cal BP. *Radiocarbon* 55:1869–1887.

Rosales D., Vidal E., Ishitsuka J., and Benavente S. 2008. Geomagnetic study of Carancas meteorite and its crater (abstract #1744). 39th Lunar and Planetary Science Conference. CD-ROM.

Scott A. C. 2000. The pre-Quaternary history of fire. *Palaeogeography, Palaeoclimatology, Palaeoecology*. 164:281–329. https://doi.org/10.1016/S0031-0182(00)00192-9.

Sibul I., Plado J., and Jõeleht A. 2017. Ground-penetrating radar and electrical resistivity tomography for mapping bedrock topography and fracture zones: a case study in Viru-Nigula, NE Estonia. *Estonian Journal of Earth Sciences* 66:142–151.

Spencer L. J. 1938. The Kaalijarv meteorite from the Estonian craters. *Mineralogical Magazine* 25:75–80.

Steier P., Dellinger F., Kutschera W., Rom W., and Wild E. M. 2004. Pushing the precision of ^{14}C measurements with AMS. *Radiocarbon* 46:5–16.

Stroeven A. P., Haettestrand C., Kleman J., Heyman J., Fabel D., Fredin O., Goodfellow B. W., Harbor J. M., Jansen J. D., Olsen L., Caffee M. W., Fink D., Lundqvist J., Rosqvist G. C., Stroemberg B., Krister N., and Jansson K. N. 2016. Deglaciation of Fennoscandia. *Quaternary Science Reviews* 147:91–121. https://doi.org/10.1016/j.quascirev.2015.09.016.

Stuiver M. and Polach H. A. 1977. Reporting of 14C data. *Radiocarbon* 19:355–363.

Szokaluk M., Jagodziński R., Muszyński A., and Szczuciński W. 2015. Ejecta blanket from the Morasko meteorite impact — First results, Bridging the Gap III: Impact Cratering. In Nature, Experiments, and Modeling, 2015 University of Freiburg, Germany, LPI contribution no. 1861, s.1100.

Szokaluk M., Jagodzinski R., Muszynski A., and Szczucinski W. 2019. Geology of the Morasko craters, Poznan, Poland —Small impact craters in unconsolidated sediments. *Meteoritics & Planetary Science* 54:1478–1494. https://doi.org/10.1111/maps.13290.

Tancredi G., Ishitsuka J., Rosales D., Vidal E., Dalmau A., Pavel D., Benavente S., Mirando P., Pereira G., Vallejos V., Varela M. E., Brandstätter F., Schultz P., Harris R. S., and Sanchez L. 2008. What do we know about the "Carancas-Desaguadero" fireball, meteorite and impact crater? (abstract #1216). 39th Lunar and Planetary Science Conference. CD-ROM.

Taylor S. R. and Mclennan S. M. 1985. *The continental crust: Its composition and evolution*. Oxford: Blackwell Scientific. 384 pp.

Wallace M. W., Gostin V. A., and Keays R. R. 1990. Acraman impact ejecta and host shales: Evidence for low-temperature mobilization of iridium and other platinoids. *Geology* 18:132–135.

Wedepohl K. H. 1995. The composition of the continental crust. *Geochimica Cosmochimica Acta* 59:1217–1232.

Wild E. M., Steier P., Fischer P., and Hoflmayer F. 2013. ^{14}C dating of humic acids from bronze and iron age plant remains from the Eastern Mediterranean. *Radiocarbon* 55:599–607.

Zazula G. D., Froese D. G., Elias S. A., Kuzmina S., and Mathewes R. W. 2007. Arctic ground squirrels of the mammoth-steppe: paleoecology of Late Pleistocene middens (~24 000–29 450 ^{14}C yr BP), Yukon Territory, Canada. *Quaternary Science Reviews* 26:979–1003. https://doi.org/10.1016/j.quascirev.2006.12.006.

19455100, 2020, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/maps.13431, Wiley Online Library on [03/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License